

Institute of Actuaries of India

Subject CT4 – Models

March 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) A stochastic model is one that recognises the random nature of the input components.

Stochastic models have the following advantages over deterministic models:

1. To reflect reality as accurately as possible, the model should imitate the random nature of the variables involved.
2. A stochastic model can provide information about the distribution of the results (eg probabilities, variances etc), not just a single best estimate figure.
3. Stochastic models allow you to use Monte Carlo simulation, which is an extremely powerful technique for solving complex problems.

(3)

ii)

For a time-inhomogenous model the transition rates are functions of t . It is certainly possible to improve the fit by using a time-inhomogenous model in this instance.

However, If the age profile is represented by a density function $f(a)$; then the overall average rate at which a healthy employee falls sick is $= \int f(a)\sigma(a)da$, roughly constant for all t . The same of course applies to the overall average rate of recovery.

(3)

iii) Proof:

Given that Increment $X_{t+u} - X_t$ for every $u > 0$ are independent of past values of X_m and non-overlapping.

Therefore, $Y = P[X_t = a \mid X_{S_1} = x_1, X_{S_2} = x_2 \dots X_{S_n} = x_n, X_s = x]$

Because $X_s = x \rightarrow X_t = X_t - X_s + x$ --- (i)

$Y = P[X_t - X_s + x = a \mid X_{S_1} = x_1, X_{S_2} = x_2 \dots X_{S_n} = x_n, X_s = x]$

Given $[X_t - X_s]$ are independent of past values of X , therefore $X_t - X_s + x$ will also be independent of past values of X_m .

Therefore:

$Y = P[X_t - X_s + x = a \mid X_s = x]$

$Y = P[X_t = a \mid X_s = x];$ Because $X_s = x$

Therefore Independent Increments satisfies Markov property

(3)

[9 Marks]

Solution 2:

- i) The probability that a mosquito will survive for 15 days, ${}_{15}P_0$, is given by the formula

$${}_{15}P_0 = \exp\left(-\int_0^{15} \mu_x dx\right)$$

Since the force of mortality is constant up to age 30 days at a value of 0.03

$$\begin{aligned} {}_{15}P_0 &= \exp\left(-\int_0^{15} 0.03 dx\right) \\ &= \exp(-[0.03x]_0^{15}) \\ &= \exp(-0.45) \\ &= 0.6376 \end{aligned}$$

(2)

- ii) Calculation of λ value:

It is given that 95% of mosquitoes are expected to have died within 90 days of age.

$$\text{Hence, } {}_{90}P_0 = {}_{30}P_0 \cdot {}_{60}P_{30}$$

$$\text{i.e., } {}_{30}P_0 \cdot {}_{60}P_{30} = 0.05$$

Thus,

$${}_{60}P_{30} = 0.05/{}_{30}P_0 \text{ -----equation}$$

(1)

$$\begin{aligned} \text{Where } {}_{30}P_0 &= \exp\left(-\int_0^{30} 0.03 dx\right) \\ &= \exp(-[0.03x]_0^{30}) \\ &= \exp(-0.90) \\ &= 0.40657 \text{ -----equation} \end{aligned}$$

(2)

From equation 1 and 2 above,

$${}_{60}P_{30} = 0.05/0.40657 = 0.12298 \text{ -----equation}$$

(3)

Also, the probability that a mosquito 30 days old will survive for a further 60 days (i.e. to exact age 90 days) is given by

$${}_{60}P_{30} = \exp\left(-\int_0^{60} \mu_{30+x} dx\right) \text{ -----equation}$$

(4)

Thus, from equation 3 & 4 above

$$\exp\left(-\int_0^{60} \mu_{30+x} dx\right) = 0.12298; \text{ where } \mu_{30+x} = -0.5 + \lambda x$$

This implies,

$$\exp\left(-\int_0^{60} \mu_{30+x} dx\right) = \exp\left(-\int_0^{60} (-0.5 + \lambda x) dx\right) = 0.12298$$

This implies,

$$\exp\{-[-0.5x + \lambda x^2/2]_0^{60}\} = 0.12298$$

$$\text{i.e. } \exp(-[-0.5 \times 60 + \lambda \times 60^2/2]) = 0.12298$$

$$\text{Taking log, i.e. } [-30 + \lambda \times 1800] = -\ln(0.12298) = 2.09573$$

$$\text{Hence, } \lambda = (32.09573 + 3)/1800$$

$$= 0.01783$$

(5)

[7 Marks]

Solution 3:

- i) Markov assumption states that the probabilities that a life at any given age will be found in any given state at any subsequent age depend only on the ages involved and the state currently occupied.

We assume that for any two distinct states i and j , and $t \geq 0$:

$${}_h p_{x+t}^{ij} = h\mu_{x+t}^{ij} + o(h)$$

and the probability that a life makes two or more transitions in a short time interval of length h is $o(h)$.

So we can write:

$${}_h p_{x+t}^{21} = h\mu_{x+t}^{21} + o(h),$$

$${}_h p_{x+t}^{31} = h\mu_{x+t}^{31} + o(h),$$

$${}_h p_{x+t}^{11} = 1 - {}_h p_{x+t}^{12} - {}_h p_{x+t}^{13} - {}_h p_{x+t}^{14},$$

$${}_h p_{x+t}^{11} = 1 - h\mu_{x+t}^{12} + h\mu_{x+t}^{14} + o(h) \text{ ; because } {}_h p_{x+t}^{13} = 0$$

Consider the interval from age x to age $x + t + h$. Using the Markov assumption we can write:

$${}_{t+h} p_x^{21} = {}_t p_x^{21} {}_h p_{x+t}^{11} + {}_t p_x^{22} {}_h p_{x+t}^{21} + {}_t p_x^{23} {}_h p_{x+t}^{31} ; \quad (\text{Kolmogorov equation})$$

Substituting ${}_h p_{x+t}^{ij}$ from above in this equation we get:

$${}_{t+h} p_x^{21} = {}_t p_x^{21} (1 - h\mu_{x+t}^{12} + h\mu_{x+t}^{14}) + {}_t p_x^{22} h\mu_{x+t}^{21} + {}_t p_x^{23} h\mu_{x+t}^{31} + o(h)$$

Rearranging this, we get

$$\frac{{}_{t+h} p_x^{21} - {}_t p_x^{21}}{h} = {}_t p_x^{22} \mu_{x+t}^{21} + {}_t p_x^{23} \mu_{x+t}^{31} - {}_t p_x^{21} (\mu_{x+t}^{12} + \mu_{x+t}^{14}) + \frac{o(h)}{h}$$

Finally, letting $h \rightarrow 0$, we obtain the result

$$\frac{\partial}{\partial t} {}_t p_x^{21} = {}_t p_x^{22} \mu_{x+t}^{21} + {}_t p_x^{23} \mu_{x+t}^{31} - {}_t p_x^{21} (\mu_{x+t}^{12} + \mu_{x+t}^{14}) ; \text{ Since } \lim_{h \rightarrow 0} \frac{o(h)}{h} = 0$$

(5)

- ii) If the life is in State 2 at time t (the probability of this is ${}_t p_x^{22}$, then to get into State 3 at age $x + t$, he must instantaneously go from State 2 to State 3 at age $x + t$ with probability μ_{x+t}^{23}

If the life is in State 3 at time t (probability ${}_t p_x^{23}$), then he must stay there. We need the $-\mu_{x+t}^{31}$ term to ensure that he doesn't move to State 1 at age $x + t$, and the $-\mu_{x+t}^{34}$ to ensure that he doesn't move to State 4. Therefore :

$$\frac{\partial}{\partial t} {}_t p_x^{23} = {}_t p_x^{22} \mu_{x+t}^{23} - {}_t p_x^{23} (\mu_{x+t}^{31} + \mu_{x+t}^{34})$$

With similar reasoning:

$$\frac{\partial}{\partial t} {}_t p_x^{32} = {}_t p_x^{31} \mu_{x+t}^{12} - {}_t p_x^{32} (\mu_{x+t}^{21} + \mu_{x+t}^{23} + \mu_{x+t}^{24})$$

(3)

[8 Marks]

Solution 4:

- i) Type I censoring: If the censoring times are known in advance (a degenerate case of random censoring) then the mechanism is called Type I censoring.

Type II censoring: If observation is continued until a predetermined number of deaths has occurred, then Type II censoring is said to be present. This can simplify the analysis, because then the number of events of interest is non-random.

(2)

- ii) The types of censoring present in the data are as below:

- i) Type I (Right) censoring of trainee who are not qualified till 30 days
- ii) Random censoring of trainee who left without qualifying within 30 days

(2)

- iii) The data can be re-arranged as shown below:

Candidate	Day	Event
F	2	Qualified
K	6	Left
A	10	Left
D	13	Qualified
H	13	Qualified
J	15	Qualified
C	18	Qualified
G	18	Left
B	20	Left
L	24	Qualified
E	30	Left
I	30	Left

The Kaplan-Meier Estimate is $S(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right)$

T_j	N_j	d_j	C_j	d_j / N_j	$1 - d_j / N_j$
0	12	0	0		
2	12	1	2	1/12	11/12
13	9	2	0	2/9	7/9
15	7	1	0	1/7	6/7
18	6	1	2	1/6	5/6
24	3	1	0	1/3	2/3

Then the Kaplan-Meier estimation of the non-qualification function is

t	S(t)
$0 \leq t < 2$	1.000
$2 \leq t < 13$	0.9167
$13 \leq t < 15$	0.7129
$15 \leq t < 18$	0.6111
$18 \leq t < 24$	0.5093
$24 \leq t < 30$	0.3395

(5)

iv)

Uses:

It is commonly used to compare the lifetime distribution of two or more population.

Limitations:

- Mainly descriptive
- Only those at risk at the observed time $\{t_j\}$ contribute to the estimate
- Cannot accommodate time dependent variables
- It is constant after the last duration at which an event is observed to occur
- It is not defined at durations longer than the duration at the last censored observation

(2)

[11 Marks]

Solution 5:

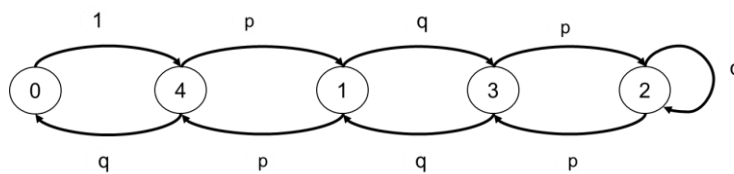
- i) Consider a Markov chain taking values in the set $S = \{i : i = 0, 1, 2, 3, 4\}$, where i represents the number of umbrellas in the place where the Actuary currently is (at home or office).

If $i = 1$ and it rains then he take the umbrella, move to the other place, where there are already 3 umbrellas, and, including the one he brings, therefore he will now have 4 umbrellas. Thus, $p_{1,4} = p$, because p is the probability of rain.

If $i = 1$ but does not rain then he do not take the umbrella, goes to the other place and find 3 umbrellas. Thus, $p_{1,3} = 1 - p \equiv q$.

However, if $i=0$, he must move to other place where 4 umbrellas are kept with probability $P_{0,4}=1$

Similarly for other states. The process is depicted by the following diagram.



(3)

ii)

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & q & p \\ 0 & 0 & q & p & 0 \\ 0 & q & p & 0 & 0 \\ q & p & 0 & 0 & 0 \end{pmatrix}$$

(2)

- iii) For stationary distribution $\pi P = \pi$

Writing the equations:

$$\begin{aligned} \pi(0) &= \pi(4).q & \text{----1} \\ \pi(1) &= \pi(3).q + \pi(4).p & \text{----2} \\ \pi(2) &= \pi(2).q + \pi(3).p & \text{----3} \\ \pi(3) &= \pi(1).q + \pi(2).p & \text{----4} \\ \pi(4) &= \pi(0) + \pi(1).p & \text{----5} \end{aligned}$$

Substituting 1 in 5

$$\begin{aligned} \pi(4) &= \pi(4).q + \pi(1).p \Rightarrow \pi(1) = \pi(4). (1-q)/p \\ \Rightarrow \pi(1) &= \pi(4). p/p \Rightarrow \pi(1) = \pi(4) & \text{--- 6} \end{aligned}$$

Substituting 6 in 2

$$\begin{aligned}\pi(1) &= \pi(3).q + \pi(4).p \Rightarrow \pi(4) = \pi(3).q + \pi(4).p \\ \Rightarrow \pi(3) &= \pi(4).(1-p)/q \Rightarrow \pi(3) = \pi(4) \quad \text{---7}\end{aligned}$$

Substituting 7 in 3

$$\begin{aligned}\pi(2) &= \pi(2).q + \pi(3).p \Rightarrow \pi(2) = \pi(3).p/(1-q) \\ \Rightarrow \pi(2) &= \pi(3).p/(1-q) \Rightarrow \pi(2) = \pi(3). \quad \text{----8}\end{aligned}$$

Substituting 6 & 7 in 4

$$\begin{aligned}\pi(3) &= \pi(1).q + \pi(2).p \Rightarrow \pi(4) = \pi(1).q + \pi(4).p \\ \Rightarrow \pi(1) &= \pi(4).(1-p)/q \Rightarrow \pi(1) = \pi(4)\end{aligned}$$

Therefore $\pi(1) = \pi(2) = \pi(3) = \pi(4) = \pi(0)/q$

But $\pi(0) + \pi(1) + \pi(2) + \pi(3) + \pi(4) = 1$, substituting in terms of $\pi(4)$
 $\pi(4).q + 4\pi(4) = 1$

$$\text{Therefore } \pi(4) = \frac{1}{4+q} = \pi(1) = \pi(2) = \pi(3) \text{ and } \pi(0) = \frac{q}{4+q}$$

(4)

- iv) He will wet every time he happens to be in state 0 and it rains. The chance he is in state 0 is $\pi(0)$. The chance it rains is p . Hence Probability that he gets wet.

$P(\text{WET})$ = Probability of being in state 0 and its raining

$$P(\text{WET}) = \frac{q}{4+q} \cdot p$$

(2)

- v) $P(\text{WET}) = 0.6 \cdot 0.4 / (4 + 0.4) = 5.45\%$

If he want the chance to be less than 1% then, clearly, he need more umbrellas. So, suppose he need N umbrellas. Set up the Markov chain as above and generalising, it is clear that

$$\pi(1) = \pi(2) = \pi(3) = \pi(4) \dots = \pi(N) = \pi(0)/q.$$

But $\pi(0) + \pi(1) + \pi(2) + \pi(3) + \pi(4) \dots \pi(N) = 1$, substituting in terms of $\pi(N)$

$$\Rightarrow \pi(N).q + N\pi(N) = 1$$

$$\Rightarrow \pi(N) = \frac{1}{N+q}$$

$$\text{Therefore probability of getting Wet } P(\text{WET}) = p \cdot \pi(0) = \frac{p \cdot q}{N+q}$$

$$\text{For } P(\text{WET}) < 1/100 \Rightarrow \frac{p \cdot q}{N+q} < 1/100$$

$$N > 100 \cdot p \cdot q - q; \text{ given } p = 0.6, q = 0.4$$

$$N > 24 - 0.4 = 23.6$$

Thus it is not worth to keep 24 umbrellas instead of 4 to reduce the probability of getting wet from 6% to 1%. On the days he did not take the umbrella but it starts raining he may buy a cheap (use and throw) umbrella from nearby local shop or take a cab/ other mode of transport to office or borrow/ share an umbrella from some friend/ colleague using the same route or wait for the rain to subside etc. (any other suitable alternative is fine).

(4)

[15 Marks]**Solution 6:**

- i) The baseline hazard applies to female patients who are given existing treatment.

(1)

- ii) The hazard for a female patient (f) is:

$$h_f(t) = h_0(t) \cdot \exp(0 + \beta_2 z_2)$$

and the hazard for a male patient (m) is:

$$h_m(t) = h_0(t) \cdot \exp(\beta_1 + \beta_2 z_2)$$

First we have to determine parameter β_1 and β_2 from the information provided.

Using $\hat{\beta}_i$ to denote our estimate of β_i , we know from i) that, if the model is correct,

$$h_m(t) = 1.05 \times h_f(t), \text{ so that:}$$

$$h_0(t) \times \exp(\hat{\beta}_1 + \hat{\beta}_2 \times z_2) = 1.05 \times h_0(t) \times \exp(\hat{\beta}_2 \times z_2)$$

$$\text{i.e., } \exp(\hat{\beta}_1) = 1.05$$

$$\text{i.e., } \hat{\beta}_1 = \ln 1.05 = 0.04879$$

And similarly, from ii), we know that:

$$h_0(t) \times \exp(\hat{\beta}_1 z_1 + 0) = 1.10 \times h_0(t) \times \exp(\hat{\beta}_1 z_1 + \hat{\beta}_2 z_2)$$

$$\text{i.e., } 1 = 1.10 \times \exp(\hat{\beta}_2 \times 1)$$

Hence,

$$\hat{\beta}_2 = \ln(1/1.10) = -0.0953$$

The hazard for a male (m) patient who has been given the new treatment (n) is:

$$h_{m,n}(t) = h_0(t) \times \exp(\hat{\beta}_1 \times 1 + \hat{\beta}_2 \times 1)$$

$$= h_0(t) \times \exp(0.04879 - 0.0953)$$

$$= h_0(t) \times \exp(-0.0465)$$

$$=0.9545 \times h_0(t)$$

Since the hazard for a female patient given the existing treatment is the baseline hazard.

The ratio of the hazard for a male patient who has been given the new treatment to that for a female patient given the existing treatment is:

$$h_{m,n}(t) / h_0(t) = 0.9545$$

Hence, the risk of death for a male patient who has been given the new treatment is less than that for a female patient given the existing treatment.

(4)

- iii) Since we know from the information given that the female whom existing treatment given had the median recovery time of 3 years i.e. $S(3) = 0.5$

This implies,

$$0.5 = \exp \left\{ - \int_{s=0}^3 h_0(s) ds \right\}$$

$$\ln(0.5) = - \int_{s=0}^3 h_0(s) ds$$

$$- \int_{s=0}^3 h_0(s) ds = -0.6931$$

For male (m) life given new treatment (n), the probability of death within 3 years is given by:

$$1 - S_{m,n}(3) = 1 - \exp \left\{ - \int_{s=0}^3 h_{m,n}(s) ds \right\}$$

$$= 1 - \exp \left\{ - \int_{s=0}^3 0.9545 * h_0(s) ds \right\}$$

$$= 1 - \exp \left\{ 0.9545 * \left(- \int_{s=0}^3 h_0(s) ds \right) \right\}$$

$$= 1 - \exp \left\{ (0.9545) * (-0.6931) \right\}$$

$$= 1 - 0.5160$$

$$= 0.4839$$

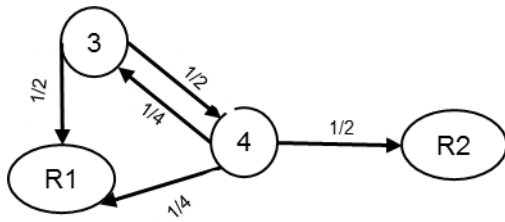
The probability of death within 3 years is 0.4839

(3)

[8 Marks]

Solution 7:

- i) The reduced state diagram is as follows. As 3 cannot be reached if one start from R1 or R2, the process is not irreducible.



(2)

- ii) The states R1 and R2 are absorbing types and once a process reached either R1 or R2 it cannot move to any other state. The Process will still continue to transition within states of R1 or R2 accordingly.

(1)

- iii) Let a_i be probability of reaching R_i and staying there forever from any other state. Therefore $a_{R1} = 1$ and $a_{R2} = 0$

$$a_i = \sum_k a_k p_{ik}, \text{ for all } i \in S$$

$$a_3 = \frac{1}{2}a_{R1} + \frac{1}{2}a_4,$$

$$a_3 = \frac{1}{2} + \frac{1}{2}a_4$$

Similarly

$$a_4 = \frac{1}{4}a_{R1} + \frac{1}{4}a_3 + \frac{1}{2}a_{R2},$$

$$a_4 = \frac{1}{4} + \frac{1}{4}a_3$$

Solving these equations

$$a_3 = \frac{5}{7} \text{ and } a_4 = \frac{3}{7}$$

Therefore if $X_0 = 3$, the chain will end up in R1 with probability $a_3 = \frac{5}{7}$

(3)

- iv) Let T be the first time the chain visits R1 or R2, for all $i \in S$, define

$$t_i = E[T | X_0 = i]$$

Therefore $t_{R1} = t_{R2} = 0$, to find t_3 and t_4 use the following equation.

$$t_i = 1 + \sum_k t_k p_{ik}, \text{ for all } i = 3, 4$$

$$t_3 = 1 + \frac{1}{2}t_{R1} + \frac{1}{2}t_4,$$

$$t_3 = 1 + \frac{1}{2}t_4$$

Similarly,

$$t_4 = 1 + \frac{1}{4}t_{R1} + \frac{1}{4}t_3 + \frac{1}{2}t_{R2},$$

$$t_4 = 1 + \frac{1}{4}t_3$$

Solving the two equations, we obtain $t_3 = \frac{12}{7}$ and $t_4 = \frac{10}{7}$

Therefore if $X_0 = 3$, the chain will end up in R1 or R2 after $t_3 = \frac{12}{7}$

(3)

[9 Marks]

Solution 8:

- i) The principle of correspondence states that the death data and the exposed to risk must be defined consistently, i.e. the numerator d_x and denominator E_x^c must correspond.

(1)

- ii) The age definition of exposed to risk to correspond to the deaths data, i.e. age next birthday.

The exposed to risk at age x next birthday may be approximated using the census approximation.

$$E_x^c = \int_{t=0}^2 P_{x,t} dx$$

Assuming the population varies linearly between census dates.

Using the trapezium rule the exposed to risk may be evaluated as

$$E_x^c = 1/2(P_{x,1/14} + P_{x,1/15}) + 1/4(P_{x,1/15} + P_{x,1/7/15}) + 1/4(P_{x,1/7/15} + P_{x,1/16})$$

$$= 1/2P_{x,1/14} + 3/4P_{x,1/15} + 1/2P_{x,1/7/15} + 1/4P_{x,1/16}$$

where $P_{x,t}$ is the population aged x next birthday at time t.

But, in this case, we have data on an age last birthday basis.

If, $P'_{x,t}$ is the population aged x last birthday at time t, then

$P_{x,t} = P'_{x-1,t}$ and the exposed to risk becomes

$$E_x^c = 1/2P'_{x-1,1/14} + 3/4P'_{x-1,1/15} + 1/2P'_{x-1,1/7/15} + 1/4P'_{x-1,1/16}$$

So, using the data given, the exposed to risk we need at age 56 is
 $E_x^c = 1/2*(50000)+3/4*(55000)+1/2*(60000)+1/4*(62000) = 111750$

and the estimated force of mortality at age 56 next birthday is

$$\mu'_{56} = (1150+1225)/111750 = 0.02125$$

(5)

iii) The implication of wrong intimation of policies in force data:

- The modified exposed to risk is 108750 and the revised force of mortality (μ'_{56}) would be 0.02184
- The actual mortality is higher due to lower exposed to risk as against the mortality used for pricing
- For the pure protection product the premium rates are highly sensitive to mortality rates and hence any under loading of mortality has huge impact on premium rates
- This may lead to under pricing and may result into financial loss to the company especially if the term of the product is long and the premiums are guaranteed
- Further, assuming the population varies linearly between census dates may also result into higher exposed to risk as in practice the large number of policies are taken at the end of tax year which may result into lower exposed to risk
- This may also result into under pricing and result into financial loss especially if large portfolio of business is written.

(3)

[9 Marks]

Solution 9: i)

a) The likelihood, L for the i th life reflects:

- The probability of the life remaining in the healthy state for total time v_i and in the sick state for time w_i , giving the factors $e^{-(\mu_{12}+\mu_{13})v_i}$ and $e^{-(\mu_{21}+\mu_{23})w_i}$ respectively.
-
- The probability of the life making the relevant number of transitions between states

giving factors $\mu_{12}^{d_{12}}, \mu_{21}^{d_{21}}, \mu_{13}^{d_{13}}$ and $\mu_{23}^{d_{23}}$

The Likelihood function is proportional to

$$L = \prod e^{-(\mu_{12}+\mu_{13})v} e^{-(\mu_{21}+\mu_{23})w} \mu_{12}^{d_{12}} \mu_{21}^{d_{21}} \mu_{13}^{d_{13}} \mu_{23}^{d_{23}}$$

Where product is taken over for all lives

$$L = e^{-(\mu_{12} + \mu_{13}) \sum v} e^{-(\mu_{21} + \mu_{23}) \sum \omega} \mu_{12}^{\sum d_{12}} \mu_{21}^{\sum d_{21}} \mu_{13}^{\sum d_{13}} \mu_{23}^{\sum d_{23}}$$

and $\sum d_{ij}$ is the number of transitions observed from state i to state j .

$\sum v$ is the total time spent in healthy state for all lives and $\sum w$ is the total time spent in sick state for all lives

(2)

b) Taking the logarithm of the likelihood in part a.

$$\text{Log } L = -(\mu_{12} + \mu_{13}) \sum v - (\mu_{21} + \mu_{23}) \sum \omega + \sum d_{12} \text{Log} \mu_{12} + \sum d_{21} \text{Log} \mu_{21} + \sum d_{13} \text{Log} \mu_{13} + \sum d_{23} \text{Log} \mu_{23}$$

Differentiating w.r.t μ_{23} ,

$$\frac{d \text{Log} L}{d \mu_{23}} = -\omega + \frac{\sum d_{23}}{\mu_{23}}$$

Setting this to 0 we obtain the maximum likelihood estimator of μ_{23} ,

$$\widehat{\mu}_{23} = \frac{\sum d_{23}}{\omega}$$

This is a maximum because $\frac{d^2 \text{Log} L}{(d \mu_{23})^2} = \frac{-\sum d_{23}}{\mu_{23}^2}$, which is always negative.

(3)

c) Maximum likelihood estimate of $\mu_{23} = \frac{700}{1600} = 0.4375$

- The maximum likelihood estimator of μ_{23} has a variance equal to $\frac{\mu_{23}}{E(V)}$, μ_{23} is the true transition rate in the population and $E[V]$ is the expected waiting time in the Sick state

-

- Approximating μ_{23} by $\widehat{\mu}_{23}$ and $E[V]$ by $\sum w$, we estimate for the variance as
 $= \frac{0.4375}{1600} = 0.000273$

At 95 per cent confidence interval around our estimate of μ_{23} is therefore

$$0.4375 \pm 1.96 \sqrt{0.000273} = (0.4052, 0.4699)$$

(3)

ii)

$$\text{a) } E(V_i) = \int_0^{b_i - a_i} P(\text{Life dies at time } t) * t dt + (b_i - a_i) * P[\text{Life Survives}]$$

$$= \int_0^{b_i - a_i} t * e^{-\mu t} \mu dt + (b_i - a_i) * e^{-\mu(b_i - a_i)} \mu dt \quad (b_i - a_i = 1 \text{ and } \mu = 0.20)$$

$$= 0.906346$$

$$\text{Total Waiting Time } E(V) = 2000 * 0.906346 = 1812.7$$

(2)

b)

$$\tilde{\mu} \sim N\left(\mu, \frac{\mu}{E(V)}\right) \equiv N(0.2, 0.010504^2)$$

$$P(\tilde{\mu} > 0.21) = 1 - \varphi\left(\frac{0.21 - 0.2}{0.010504}\right) = 1 - \varphi(0.9522) = 1 - .8295 = 17.05\%$$

(2)

[12 Marks]

Solution 10:

i) Methods of graduation are:

- a) Fitting a parametric formula: use the third differences of the graduated rates which should be small in magnitude and progress regularly to ensure smoothness of rates
- b) Graphical graduation: A formula with a small number of parameters is chosen, the resulting graduation will be acceptably smooth.

Sometimes hand-polishing is also necessary to ensure smoothness of rates for the above methods.

- c) Standard table: Standard table is already acceptably smooth.

(3)

ii) Reasons for smoothing

- By smoothing the experience, we can make use of data at adjacent ages to improve the estimates at each age.
- We believe that mortality varies smoothly with age. Therefore the crude estimate of mortality at any age carries information about mortality at adjacent ages.
- To remove sampling (or random) errors.
- To produce a smooth set of rates that are suitable for a particular purpose such as premium for life insurance contracts.
- Any irregularities, jumps and anomalies in financial quantities (such as premiums for life insurance contracts) are hard to justify to customers.

(2)

iii) Appropriate method of graduation

- a) The male members covered under PMJJY (Pradhan Mantri Jeevan Jyoti Yojna)

With reference to a standard table, because there are many extant tables dealing with male population covered between ages 18 & 60 years.

- b) The female population of a large developing country
By parametric formula, because the experience is large (or because the graduated rates may form a new standard table for the country).

- c) The third gender patients suffering from brain cancer in a particular region.

Graphical, because no suitable standard table is likely to exist and the experience is small.

(3)

- iv) To test for overall goodness of fit we use the χ^2 test.

The null hypothesis is that the graduated rates are the same as the true underlying rates for the block of business.

The test statistic

$$\chi_n^2 = \sum_x^{all\ ages} Z_x^2$$

where n is the degrees of freedom.

Age	Exposed to risk	Observed Death	Graduated Rates (qx)	Expected Death	Z_x	Z_x^2
50	1280	4	0.00230	2.9440	0.6155	0.3788
51	2030	5	0.00262	5.3186	-0.1381	0.0191
52	1950	11	0.00296	5.7720	2.1761	4.7353
53	2160	7	0.00331	7.1496	-0.0559	0.0031
54	2480	10	0.0037	9.1760	0.2720	0.0740
55	1455	7	0.00415	6.0383	0.3914	0.1532
56	2100	11	0.00463	9.7230	0.4095	0.1677
57	1865	17	0.00518	9.6607	2.3613	5.5757
58	1990	16	0.00578	11.5022	1.3262	1.7588
59	1725	9	0.00645	11.1263	-0.6374	0.4063
Total					6.7204	13.2720

Observed Test Statistic is 13.27

The number of age groups (n) is 10, but we lose an unknown number of degrees for the graduation, perhaps 2. So $n = 8$, say.

The critical value of the chi-squared distribution with 8 degrees of freedom at the 5% level is 15.51.

Since $13.27 < 15.51$

We do not reject the null hypothesis.

(4)
[12 Marks]
